Supporting Information

Geometry and Quantum-Chemical Corrections in Predicting Fluoride Ion Affinity

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S1. Versioned data and complete geometry preparation

File	Figshare file ID	Verified MD5
FIA49k.csv.gz	43289211	76b9018afda2f8b9c776d1555f55180f
FIA49k_predictions.csv.gz	43289322	7f0f55c3e876d34452abd7bbb0c0e657
FIA49k.zip	43289292	60e9ffd7280bdfd6baeb37d6a641f81b

All 48,986 molecules and 1,617,147 atoms are retained; maximum atom count 118. Atomic numbers: 1, 5, 6, 7, 8, 9, 13, 14, 15, 16, 17, 31, 32, 33, 35, 49, 50, 51, 52, 53, 82, 83. SHA256 fingerprints for the source archives and prepared NPZ are in reports/geometry_provenance.json.

Assignment	n
train	32333
validate	7001
test	6732
test_2	1200
test_4	911
test_3	763
test_6	31
test_5	15

All atom and target arrays are finite, coordinate row counts are exact, unknown atomic symbols raise an error, and Compound identifiers are unique. No fallback from acid to fluoride-adduct geometry is allowed. The first-atom accepting-center convention is checked against central-element identity for every record. Element-sequence agreement cannot alone exclude permutations of chemically equivalent atoms.

S2. Partition definitions and regression checks

Generic ligand-scaffold multiset: 12,448 groups containing 46,066 molecules, assigned as 33,460 training, 6,379 validation and 6,227 test records. Individual-ligand overlap with training is 80.8% in validation and 84.6% in test.

Exact complete ligand set: 43,226 groups containing 46,066 molecules, assigned as 32,184 training, 6,899 validation and 6,983 test records. Individual-ligand overlap with training is 81.1% in validation and 80.3% in test.

Canonical acid identity auditing found zero records crossing partitions under the defined RDKit canonicalization. This does not prove the absence of related ligand families, tautomeric relationships, protonation variants or near-duplicate geometry. The full canonical grouping keys and original/reassigned molecule identities are included in the machine-readable manifests. Generic scaffold keys retain the multiset of ligand scaffolds; an individual scaffold may recur in a different multiset.

The regression suite checks disjoint and reproducible legacy splits; strict prediction identity and target matching; malformed geometries and unknown elements; rotation, translation and permutation invariance; independent batching; exact checkpoint restoration; train-only target normalization; train-only electronic-feature standardization and source fingerprint validation; and deterministic ligand grouping. The validation record accompanies the code. Tests of geometric invariance cover the new invariant models; they are not a proof of physical causality.

S3. Neural configurations and seed-level test results

Run	Parameters	Epochs	Selected	Val MAE	Test MAE	Test RMSE
radial-site-s1	378,757	86	61	11.918	12.417	18.670
radial-site-s2	378,757	85	60	11.919	12.095	18.111
radial-site-s3	378,757	100	97	11.844	12.161	18.491
radial-sum-s1	345,989	99	74	11.971	12.299	18.291
radial-sum-s2	345,989	87	62	11.846	12.507	18.598
radial-sum-s3	345,989	83	58	11.873	12.373	18.716
egnn-sum-s1	758,408	100	83	11.944	12.255	18.883
egnn-sum-s2	758,408	100	99	11.699	12.013	18.285
egnn-sum-s3	758,408	87	62	11.720	11.965	18.524
dimenet-s1	1,885,830	100	94	10.496	10.993	17.010
dimenet-s2	1,885,830	100	83	10.834	11.506	17.732
dimenet-s3	1,885,830	100	99	10.583	11.119	17.276
acid-qm-s1	413,061	100	94	11.549	12.152	18.554
acid-qm-s2	413,061	93	68	11.769	12.168	18.437
acid-qm-s3	413,061	72	47	11.759	12.222	18.804
scaffold-site-s1	378,757	92	67	12.558	12.630	17.886
scaffold-site-s2	378,757	100	98	12.407	12.790	18.182
scaffold-site-s3	378,757	97	72	12.520	12.893	18.380
scaffold-qm-s1	413,061	53	28	12.322	12.762	18.155
scaffold-qm-s2	413,061	69	44	12.086	12.780	18.391
scaffold-qm-s3	413,061	72	47	12.213	12.713	18.180
composition-s1	378,757	56	31	28.397	28.695	39.451
composition-qm-s1	413,061	64	39	19.732	20.155	29.017

Scaffold runs use their separate scaffold test, not the official main test. All errors are kJ/mol. An unweighted ensemble and a seed-average MAE are different estimands and are reported separately. Runtime is observed shared-host epoch wall time, not an isolated speed comparison. CUDA index_add may produce nondeterministic rounding even with fixed RNG seeds. Full configurations, data fingerprints and history logs accompany each checkpoint.

S4. Neural ensemble errors across partitions

Model	Split	n	MAE	RMSE	R2
radial-site	test	6732	11.152	17.225	0.9481
radial-site	test_2	1200	13.398	18.432	0.9215
radial-site	test_3	763	23.104	30.382	0.8720
radial-site	test_4	911	14.454	18.618	0.9224
radial-site	test_5	15	7.615	9.583	0.8962
radial-site	test_6	31	10.389	14.811	0.9322
radial-sum	test	6732	11.222	17.222	0.9481
radial-sum	test_2	1200	13.942	19.534	0.9118
radial-sum	test_3	763	22.876	30.245	0.8732
radial-sum	test_4	911	14.143	17.939	0.9279
radial-sum	test_5	15	9.017	11.378	0.8537
radial-sum	test_6	31	9.364	13.233	0.9459
egnn-sum	test	6732	11.044	17.298	0.9477
egnn-sum	test_2	1200	13.697	19.445	0.9126
egnn-sum	test_3	763	23.544	31.357	0.8637
egnn-sum	test_4	911	13.438	17.616	0.9305
egnn-sum	test_5	15	3.260	5.018	0.9715
egnn-sum	test_6	31	9.018	12.131	0.9545
dimenet	test	6732	10.288	16.394	0.9530
dimenet	test_2	1200	11.931	16.809	0.9347
dimenet	test_3	763	20.760	27.895	0.8921
dimenet	test_4	911	14.726	19.261	0.9169
dimenet	test_5	15	6.392	7.609	0.9346
dimenet	test_6	31	4.394	6.008	0.9888
acid-qm	test	6732	11.125	17.423	0.9469
acid-qm	test_2	1200	13.478	18.347	0.9222
acid-qm	test_3	763	19.887	26.055	0.9059
acid-qm	test_4	911	13.877	18.294	0.9250
acid-qm	test_5	15	6.067	8.828	0.9119
acid-qm	test_6	31	10.836	16.921	0.9116
scaffold-site	test	6227	11.660	16.828	0.9446
scaffold-qm	test	6227	11.705	17.009	0.9434
composition	test	6732	28.695	39.451	0.7277
composition	test_2	1200	37.944	48.967	0.4457

Model	Split	n	MAE	RMSE	R2
composition	test_3	763	33.981	43.911	0.7326
composition	test_4	911	25.621	33.465	0.7492
composition	test_5	15	36.785	40.195	-0.8258
composition	test_6	31	25.230	33.265	0.6582
composition-qm	test	6732	20.155	29.017	0.8527
composition-qm	test_2	1200	25.481	34.479	0.7252
composition-qm	test_3	763	22.842	30.332	0.8724
composition-qm	test_4	911	21.049	29.001	0.8116
composition-qm	test_5	15	18.522	20.596	0.5206
composition-qm	test_6	31	21.827	33.458	0.6542

S5. Classical baselines and quantum-refinement controls

qm_baseline

Model	Split	n	MAE	RMSE
metadata	test	6732	37.569	49.465
metadata	test_2	1200	51.748	62.948
metadata	test_3	763	54.034	63.876
metadata	test_4	911	39.702	48.455
metadata	test_5	15	42.694	47.346
metadata	test_6	31	81.991	92.345
metadata_plus_LA_QM	test	6732	23.020	31.939
metadata_plus_LA_QM	test_2	1200	29.262	38.617
metadata_plus_LA_QM	test_3	763	25.799	33.267
metadata_plus_LA_QM	test_4	911	21.361	28.366
metadata_plus_LA_QM	test_5	15	39.281	40.008
metadata_plus_LA_QM	test_6	31	39.802	59.367

rdkit_baseline

Model	Split	n	MAE	RMSE
RDKit2D	test	6732	18.729	25.821
RDKit2D	test_2	1200	23.198	30.162
RDKit2D	test_3	763	24.989	33.197
RDKit2D	test_4	911	21.162	28.056
RDKit2D	test_5	15	27.060	31.027
RDKit2D	test_6	31	9.288	12.577
RDKit2D_plus_LA_QM	test	6732	17.347	24.176
RDKit2D_plus_LA_QM	test_2	1200	22.978	29.944
RDKit2D_plus_LA_QM	test_3	763	21.848	29.331
RDKit2D_plus_LA_QM	test_4	911	18.034	24.415
RDKit2D_plus_LA_QM	test_5	15	25.807	28.398
RDKit2D_plus_LA_QM	test_6	31	21.501	30.282

refinement

Model	Split	n	MAE	RMSE
PBEh3c_raw	test	6732	23.332	26.268
PBEh3c_raw	test_2	1200	26.732	29.532
PBEh3c_raw	test_3	763	29.759	32.182
PBEh3c_raw	test_4	911	26.639	29.333
PBEh3c_raw	test_5	15	14.195	15.127
PBEh3c_raw	test_6	31	14.635	16.728
PBEh3c_global_bias	test	6732	10.062	12.616
PBEh3c_global_bias	test_2	1200	10.674	13.322
PBEh3c_global_bias	test_3	763	11.307	14.129
PBEh3c_global_bias	test_4	911	9.999	12.945
PBEh3c_global_bias	test_5	15	9.783	10.195
PBEh3c_global_bias	test_6	31	9.739	11.753
PBEh3c_class_bias	test	6732	7.149	9.218
PBEh3c_class_bias	test_2	1200	8.407	10.854
PBEh3c_class_bias	test_3	763	8.332	10.328
PBEh3c_class_bias	test_4	911	6.821	8.748
PBEh3c_class_bias	test_5	15	3.231	5.589

Model	Split	n	MAE	RMSE
PBEh3c_class_bias	test_6	31	6.818	8.613
delta_metadata	test	6732	4.845	6.314
delta_metadata	test_2	1200	6.938	8.734
delta_metadata	test_3	763	5.000	6.492
delta_metadata	test_4	911	3.789	5.026
delta_metadata	test_5	15	5.371	6.428
delta_metadata	test_6	31	3.208	4.080
delta_RDKit_LA_QM	test	6732	3.289	4.485
delta_RDKit_LA_QM	test_2	1200	3.765	4.897
delta_RDKit_LA_QM	test_3	763	4.034	5.193
delta_RDKit_LA_QM	test_4	911	3.030	4.092
delta_RDKit_LA_QM	test_5	15	4.298	4.773
delta_RDKit_LA_QM	test_6	31	2.323	2.832

grouped_scaffold

Model	Split	n	MAE	RMSE
RDKit2D	test	6227	19.095	26.180
delta_RDKit_LA_QM	test	6227	3.348	4.453

The final HGB stage budget is matched across primary, grouped and refinement experiments: stages 100/400/1000 are selected on validation only. Every fitted feature list and selection score is saved in the corresponding config.json. The lower-level FIA in refinement is an explicit adduct-dependent input; it is not included in the acid-QM fusion models.

S6. Paired uncertainty and subgroup interpretation

Model	Split	Δ MAE	95% lower	95% upper
radial-site	test	-0.960	-1.194	-0.729
radial-site	test_2	-1.041	-1.731	-0.365
radial-site	test_3	-0.977	-2.054	0.116
radial-site	test_4	-3.480	-4.289	-2.663
radial-site	test_5	-3.308	-7.129	1.392
radial-site	test_6	1.858	0.886	2.855
radial-sum	test	-0.890	-1.121	-0.658
radial-sum	test_2	-0.498	-1.204	0.238
radial-sum	test_3	-1.204	-2.321	-0.087
radial-sum	test_4	-3.791	-4.625	-2.988
radial-sum	test_5	-1.905	-5.902	3.183
radial-sum	test_6	0.834	-0.379	2.031
egnn-sum	test	-1.068	-1.313	-0.837
egnn-sum	test_2	-0.742	-1.422	-0.043
egnn-sum	test_3	-0.537	-1.718	0.590
egnn-sum	test_4	-4.496	-5.174	-3.811
egnn-sum	test_5	-7.662	-10.714	-4.395
egnn-sum	test_6	0.487	-0.808	1.733
dimenet	test	-1.824	-2.060	-1.592
dimenet	test_2	-2.509	-3.196	-1.808
dimenet	test_3	-3.321	-4.509	-2.179
dimenet	test_4	-3.208	-4.241	-2.252
dimenet	test_5	-4.531	-7.537	-1.209
dimenet	test_6	-4.136	-4.971	-3.299
acid-qm	test	-0.987	-1.227	-0.749
acid-qm	test_2	-0.962	-1.661	-0.258
acid-qm	test_3	-4.193	-5.474	-2.938
acid-qm	test_4	-4.057	-4.981	-3.163
acid-qm	test_5	-4.856	-8.492	-0.166
acid-qm	test_6	2.305	0.978	3.573
scaffold-site	test	-7.435	-7.809	-7.049
scaffold-qm	test	-7.390	-7.770	-7.000
composition	test	16.583	15.952	17.190
composition	test_2	23.504	21.795	25.210
composition	test_3	9.900	8.089	11.715
composition	test_4	7.687	6.420	8.983
composition	test_5	25.862	17.058	34.221

Model	Split	Δ MAE	95% lower	95% upper
composition	test_6	16.699	14.655	18.698
composition-qm	test	8.043	7.611	8.483
composition-qm	test_2	11.041	9.794	12.399
composition-qm	test_3	-1.239	-2.840	0.355
composition-qm	test_4	3.115	2.087	4.131
composition-qm	test_5	7.599	3.193	12.941
composition-qm	test_6	13.297	11.173	15.557

Δ MAE is new-model ensemble error minus reference error; negative values favor the new model. Official-partition reference: released FIA-GNN. Scaffold reference: the independently refitted RDKit HGB baseline. Each bootstrap resample preserves central-class counts and pairs predictions on the same molecules. These intervals condition on the fitted weights. The complete by_class.csv contains per-class n, target variance, MAE, RMSE and R2; small classes and low-variance targets require caution.

S7. Geometry-source and paired-structure diagnostics

Seed	Split	xTB MAE	PBEh MAE	Δ MAE	Mean |Δ prediction|
1	test	14.385	12.417	1.969	7.117
1	test_2	16.621	14.913	1.709	6.848
1	test_3	26.005	24.738	1.267	7.876
1	test_4	16.647	15.921	0.726	5.127
1	test_5	11.279	8.116	3.162	3.903
1	test_6	16.706	13.330	3.376	5.282
2	test	13.919	12.095	1.824	7.196
2	test_2	16.117	14.717	1.401	6.504
2	test_3	25.641	23.918	1.724	7.571
2	test_4	16.239	15.527	0.712	4.959
2	test_5	10.860	8.484	2.376	3.298
2	test_6	10.419	7.835	2.583	5.942
3	test	14.022	12.161	1.860	6.992
3	test_2	16.354	14.961	1.393	6.513
3	test_3	26.121	24.795	1.326	7.639
3	test_4	16.070	14.907	1.163	5.560
3	test_5	8.843	8.227	0.616	2.832
3	test_6	17.693	12.877	4.816	7.212

Paired acid/adduct structure check: N_adduct=N_acid+1; all preceding Z equal; final Z=9; xTB Z equal. Element-sequence equality checks the archive order convention; identical-element atom permutations cannot be excluded by Z alone.

The heavy-atom pair-distance RMS change is sqrt(mean[(d_ij,adduct - d_ij,acid)^2]) over unordered heavy-atom pairs. The 2.7 Å center-shell cutoff is geometric and does not assign covalent bonds. Distances are in Å. The added fluoride is the final atom in the checked archive convention. These released geometries are used for structural diagnostics, independently of any energetic deformation analysis. The full 48,986-row geometry diagnostic export is included in the archive.

S8. Matched-ligand reference-method corrections

Center A	Center B	Ligand sets	B - A	95% lower	95% upper
P(III)	As(III)	34	2.662	0.838	4.621
As(III)	Sb(III)	69	-0.808	-2.183	0.605
Sb(III)	Bi(III)	74	4.156	3.552	4.705
Si(IV)	Ge(IV)	30	2.502	0.552	4.465
Ge(IV)	Sn(IV)	22	-1.648	-3.683	0.337
Sn(IV)	Pb(IV)	10	5.413	3.202	7.480
Si(II)	Ge(II)	90	-0.083	-0.838	0.728
Ge(II)	Sn(II)	98	6.400	5.763	7.033
Sn(II)	Pb(II)	103	6.349	6.005	6.681

The correction is FIA(DSD-BLYP) minus FIA(PBEh-3c), in kJ/mol. Comparisons use only training molecules. Repeated records are averaged within each canonical ligand-set/center-class cell before pairing. The nine neighboring-center contrasts were defined before inspection and are all retained. Intervals use 2,000 bootstrap samples over matched ligand sets with seed 20261113; they are descriptive, without a multiple-testing significance claim.

S9. Reproduction and provenance

README.md gives the preparation, training, evaluation and analysis commands. reports/experiment_plan.json distinguishes the initial fixed comparisons from exploratory acid-QM and geometry-source extensions. requirements-benchmark.txt and run configurations record the software environment; THIRD_PARTY.md records upstream code commits and licenses. A frozen archive contains split identities, all prediction exports, histories and checkpoints, together with a checksum manifest. The storage manifest provides SHA256 verification for each restored file.